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SUPERSEDING
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MILITARY SPECIFICATION

TELESCOPE, M84

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of instrument known as Telescope, M84 used on .30 Caliber Rifle M1c.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS.

2.1 The following specifications and drawing of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-P-11638 —Packaging and Packing of Telescope, M84

U. S. ARMY

51-7 —Instruments, Fire-Control; General Specification Governing the Manufacture, and Inspection of

51-70-2 —Optical Components for Fire-Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of

DRAWINGS

ORDNANCE CORPS

F7667417 —Telescope, M84

(Copies of specifications, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements and associated tests common to all fire control instruments as contained in Specifications 51-7 and 51-70-2 shall be mandatory in so far as they are applicable.

3.2 Materials shall be in accordance w the drawings, general specifications and terial specifications forming part of this sp fication.

3.3 Collimation.—The following requirements shall be met when the telescope is set up as shown on the Sectional Views Drawing F7667417 with the centerline of the exterior surface of the body tube horizontal, and with the longer reticle line horizontal.

3.3.1 With the range and windage knobs set at zero, the optical axis as determined by the top mid point of the vertical reticle line shall coincide with the geometric axis of the exterior surface of the telescope body tube within plus or minus 0.5 mil.

3.3.2 The vertical reticle line shall be vertical within 1.5°. (This reticle line shall be in the lower half of the field when viewed through the eyepiece).

3.3.3 When the range or windage knob is rotated to move the reticle a maximum distance (approximately 40 min.), the deviation from the vertical or horizontal plane shall not exceed three minutes of true field.

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3.4 Eyepiece focus.—The eyepiece (fixed) focus shall be set within the range of minus 1.25 to minus .375 diopters.

3.5 Reticle travel.

3.5.1 *Elevation Accuracy* shall be as indicated in the table below:

RANGE ELEVATION TOLERANCE

Yards	Mils	Mil Plus or Minus
0	0	0
400	3.5	0.3
900	12.1	0.4

3.5.2 The windage knob shall be rotable to the right and left an amount at least equal to e maximum graduation.

.5.3 From the 900 yard knob position there be an additional clearance between the B7667465, and housing, A7631585, to al- he reticle to move downward (as viewed throu h the eyepiece) a distance equal to 1.5 mils. In a like manner, from the 20 left and right windage knob positions there shall be an additional clearance of 10 minutes. (To check these additional movements of the reticle for range and windage, unscrew the nuts A7592060 as far as possible. Pull out the knobs so that the teeth are disengaged from the detents and reset both knobs to a position sufficiently far from the stop to allow this movement. Having satisfactorily checked these measurements recollimate the telescope.)

3.5.4 *Backlash* in each of the range and windage mechanisms shall not exceed a value of one click of the knob as measured in the field of view.

3.6 Definition.—Resolving power on the optical axis for a parallel bar test target shall be 13 seconds of arc or better.

3.7 Parallax.—When viewing a target at any range between 100 and 120 yards, parallax shall not be in excess of 0.25 mil (¼ width of vertical reticle line) between the target image and the vertical reticle at the center of field.

3.8 Vibration.—After being subjected to the vibration test of Specification 51–70–2 there shall be no loose dirt in excess of the following:

3.8.1 Dirt on the adjacent surfaces of the window and objective lens shall not exceed four in number of a size not larger than a 50 dig.

3.8.2 Dirt shall not exceed one in number (on each surface) of a size not larger than a 5 dig on the second surface of the objective lens, both surfaces of the reticle, and the surface of the erector lens adjacent to the reticle.

3.8.3 Dirt between the adjacent surfaces of the erector lenses shall not exceed three in number of a size not larger than a 50 dig.

3.8.4 Dirt between the adjacent surfaces of the erector lens and eyelens shall not exceed three in number of a size not larger than a 40 dig.

3.8.5 Dirt between the adjacent surfaces of the eyelens and window shall not exceed three in number of a size not larger than a 40 dig.

3.9 Sealing.—The telescope shall show no evidence of leakage when subjected to the pressure test.

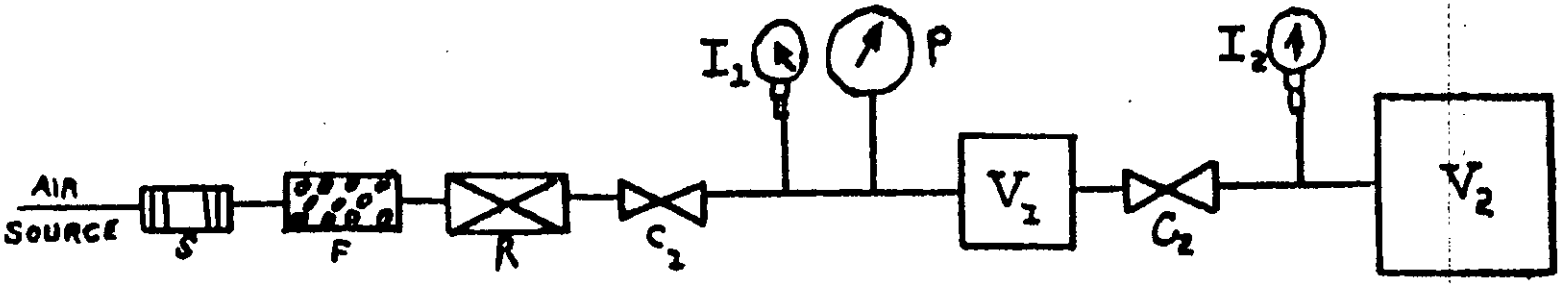
3.10 Temperature.—Temperature requirements and test Type I outlined in Specification 51–7 shall apply, except minus 65° F. shall be substituted for minus 45° F., and mins 80° F. substituted for minus 60° F.

3.11 Workmanship.—Workmanship shall be in accordance with Specifications 51–7 and 51–70–2.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Unless otherwise specified, each telescope shall be inspected for conformance with this specification and as specified in Specifications 51–7 and 51–70–2.

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S= Snap-on Plug

F= Filter

R= Air Regulator

C_1, C_2= Valves

I_1, I_2= Dial indicators

P= Pressure gage

V_1= Airtight cylinder whose internal volume is adjusted to be approximately equal to the internal air volume of instrument to be tested.

V_2= Airtight adapter whose internal volume is constructed to conform to the outside contour of instrument to be tested allowing a minimum space between the instrument and the adapter to permit ease in loading and unloading.

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4.2 **Definition Test.**—The test target shall be of the pattern illustrated in Specification 51–70–2, Figure VI–A. The images shall be viewed with the aid of a 6 power auxiliary telescope, scaled in diopters. The test pattern shall be resolved in each of the four meridians by an adjustment of the auxiliary telescope not to exceed plus or minus ⅛ diopter from the setting for best focus on the numeral in the center of the test pattern. Illumination of the test chart shall be such that the image seen through the auxiliary telescope will have an apparent brightness between 10 and 30 millilamberts.

4.3 **Pressure Test.**—The arrangement for the testing apparatus, similar to that shown on Fixture Pressure Testing F7680673 exclusive of adapter V_2 is indicated below:

4.3.1 *Procedure.*—Prior to testing any instrument the apparatus should be checked for possible leakage by opening C_1 and C_2, and injecting a pressure into the system until P reads 6 psi. Thereupon, close C_1 and C_2, and adjust I_1 and I_2 to zero. After a two minute period has elapsed, there shall be no drop in pressure as shown on either dial indicator. Exhaust system by disconnecting snap-on plug.

(1) Place instrument in V_2.
(2) Close C_2.
(3) Set and record a pressure in V_1 to a value just above 5 psi which shall be indicated on P by manipulating the air regulator to a desired pressure and opening C_1; NOTE PRESSURE LIMIT IN STEP (6) BELOW.
(4) Close C_1.
(5) Adjust I_1 to read zero.
(6) Open C_2 gradually, and note pressure now registered at P. (The pressure desired in 5 psi). Caution should be exercised to prevent the pressure after opening valve C_2 from exceeding 6 psi.
(7) If unsuccessful in obtaining this desired pressure exhaust system and repeat steps (2) and (6) until the desired pressure of 5 psi is attained.
(8) With the pressure in V_1 and V_2 at 5 psi, adjust I_2 to zero.
(9) If I_2 shows a gradual decrease within a period of two minutes, the instrument is leaking, and should be replaced by another with steps (1) to (8) repeated until the condition is reached when I_2 remains at zero.
(10) Replace the instrument with one that is known to be defective. (If in doubt, remove or loosen a screw from the instrument which will allow air to creep into its interior.)
(11) Close C_2.
(12) Open C_1 until I_1 is zero. (P should be approximately the same value as it was in step (3) for the final selected pressure.)
(13) Open C_2 gradually (P should indicate approximately 5 psi.)
(14) If I_2 reads zero, then the first and second instruments tested are defective. However, should I_2 fail to reach zero, then the first instrument is acceptable but the second is defective.

(*Note.*—In the event that the first instrument is also rejected, it should then be replaced and steps (1) to (14), inclusive, repeated until one has been found satisfactory. If, however, it has been determined that the first instrument tested is acceptable, then no further adjustments are necessary. The remaining instruments to undergo the test shall be inserted into the adapter and steps (11) to (14), inclusive, repeated in each instance.

If an extremely gradual decrease is detected and doubt exists as to whether or not the instrument leaks, it should be removed from the adapter and allowed to remain for a period of not less than 10 minutes before subjecting it to another test in order to allow the trapped air to escape.)

4.3.2 After 50 consecutive instruments have passed the sealing test and the remainder divided into evenly distributed lots of not less than 50 each, then 20% of each lot shall be tested. Failure of any one instrument to pass

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shall subject each instrument in that particular lot to undergo the test; those passing, however, shall be acceptable. Where a lot is less than 50 the entire amount shall be pressure tested.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking shall be in accordance with specification MIL–P–11688.

6. NOTES

6.1 Ordering Data.—Invitation for bids and contracts should specify the title, number, date of this specification, and type of packing.

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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